

AI/ML Engineering Roadmap+

A Complete Self-Study Path from Foundations to Professional AI Engineering

Study Order

Phase 0	Math Foundations + Python (run in parallel throughout)
Phase 1	Python · Statistics · EDA · Visualization (GFG Weeks 0–4)
Phase 2	SQL + Databases
Phase 3	Machine Learning + WQU Data Science Projects (GFG Weeks 5–10)
Phase 4	Git · Docker · CI/CD · Core MLOps
Phase 5	Deep Learning+ WQU DL Projects + Reinforcement Learning + DSA (GFG Weeks 11–16)
Phase 6	Modern AI Engineering (NLP · CV · LLMs · GenAI · RAG · Vector DBs · Agents)
Phase 7	Advanced Track — Cloud · Data Engineering · Google Professional ML Engineer

PHASE 1 Python · Statistics · EDA · Visualization

✓ Fully covered by GFG Weeks 0–4. No external course required.

- 1.1 Data Science Overview — *GFG Week 0*
 - 1.2 Python Foundations — *GFG Week 1*
 - 1.3 Statistics for Data Science — *GFG Week 2*
 - 1.4 EDA with Pandas & NumPy — *GFG Week 3*
 - 1.5 Data Visualization — Matplotlib, Seaborn, Plotly — *GFG Week 4*
- GFG Course: <https://www.geeksforgeeks.org/batch/ds-16?tab=Resources>

PHASE 0 Python + Math Foundations

Run Python (0.0) alongside every phase. Complete Math (0.1–0.4) before starting Phase 1.

0.0 — Python

- **Python Programming MOOC 2026** — *University of Helsinki · free · world-class*
<https://programming-26.mooc.fi/>

0.1 — Calculus

- **AP Calculus AB** — *Khan Academy*
<https://www.khanacademy.org/math/ap-calculus-ab>

0.2 — Linear Algebra

- **Linear Algebra** — *Khan Academy*
<https://www.khanacademy.org/math/linear-algebra>

0.3 — Statistics & Probability

- **Statistics & Probability** — *Khan Academy*

<https://www.khanacademy.org/math/statistics-probability>

Alternative Path — Mathematics for Machine Learning (if Khan Academy feels too disconnected from AI)

Mathematics for Machine Learning Specialisation — Imperial College London · Coursera · audit free · Linear Algebra, Multivariate Calculus & PCA

0.1 — Calculus

- **Multivariate Calculus** — Imperial College London
<https://www.youtube.com/watch?v=cWZLPv4ZJhE&list=PLiiljHvN6z193BBzS0Ln8NnqQmzimTW23>
- **Essence of Calculus** — 3Blue1Brown · watch first
<https://www.youtube.com/playlist?list=PLZHQObOWTQDMSr9K-rj53DwVRMYO3t5Yr>

0.2 — Linear Algebra

- **Linear Algebra for Neural Networks** — Imperial College London
https://www.youtube.com/playlist?list=PLiiljHvN6z1_o1ztXTKWPrShrMrBLo5P3
- **Essence of Linear Algebra** — 3Blue1Brown · must watch
https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab

— PCA (after linear algebra)

- **PCA Course** — Marc Peter Deisenroth · co-author of Mathematics for Machine Learning
https://www.youtube.com/playlist?list=PL93aLKqThq4iDbY_wxfh3WqxkkzqH4aoo
 - **Companion Textbook (free PDF)** — mml-book.com
<https://mml-book.com>
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PHASE 1 Python + Statistics + EDA + Visualization

These are already covered by GFG Week 0 → Week 4.

✓ 1.1 – Data Science Overview

👉 GFG Week 0

✓ 1.2 – Python Foundations

👉 GFG Week 1

✓ 1.3 – Statistics for Data Science

👉 GFG Week 2

✓ 1.4 – EDA with Pandas & NumPy

👉 GFG Week 3

✓ 1.5 – Data Visualization (Matplotlib, Seaborn, Plotly)

👉 GFG Week 4

! No extra course required here — GFG covers this perfectly.

PHASE 2 SQL + Databases

⚠ GFG does not teach SQL. Complete the resources below before Phase 3.

2.1 SQL Murder Mystery — Joon Park and Cathy He
<https://mystery.knightlab.com/>

2.2 a) SQL Track · PRIMARY — DataCamp
<https://app.datacamp.com/learn/skill-tracks/sql-fundamentals>

- 2.2.1 Introduction to SQL
 - 2.2.2 Intermediate SQL
 - 2.2.3 Joining Data in SQL
 - 2.2.4 Database Design (chapters 1-2 only, partial)
- Stopped early — course did not teach CREATE TABLE syntax

2.2 b) SQLBolt — CREATE TABLE and Database Building · PRIMARY

<https://sqlbolt.com/>

Complete lessons 13–18 only:

Lesson 13: Inserting Rows

Lesson 14: Updating Rows

- Lesson 15: Deleting Rows
- Lesson 16: Creating Tables
- Lesson 17: Altering Tables
- Lesson 18: Dropping Tables

2.3 b) SQL Track · Use when you don't have Datacamp — SQL Bot
<https://sqlbolt.com/>

2.4 SQL + Pandas Connection · PRIMARY

Learn how to connect a SQLite database to Python/pandas using sqlite3 and pd.read_sql().

<https://www.youtube.com/watch?v=pd-0G0MigUA>

OR USE THIS

STEP 1: Create your database file

```
import sqlite3 import pandas as pd
```

```
#This creates galamsey.db in your project folder
```

```
#If it already exists, it just opens it
```

```
conn = sqlite3.connect("galamsey.db")
```

That's it. Your database file now exists on your computer. Think of conn as the door between Python and your database — everything goes through it.

STEP 2: Create your tables inside it

```
conn.execute("""
    CREATE TABLE IF NOT EXISTS regions (
        region_id INTEGER PRIMARY KEY,
        region_name TEXT NOT NULL, population INTEGER
    )
""")
conn.execute("""
    CREATE TABLE IF NOT EXISTS rivers (
        river_id INTEGER PRIMARY KEY,
        river_name TEXT NOT NULL,
        region_id INTEGER REFERENCES regions(region_id)
    )
""")
```

conn.commit() # saves everything — never forget this line

STEP 3: Insert your data

```
conn.execute(""" INSERT INTO regions (region_id, region_name, population) VALUES (1, 'Ashanti', 5000000) """)
conn.execute(""" INSERT INTO regions (region_id, region_name, population) VALUES (2, 'Western', 2800000) """)
conn.execute(""" INSERT INTO rivers (river_id, river_name, region_id) VALUES (1, 'Pra River', 1) """)
conn.execute(""" INSERT INTO rivers (river_id, river_name, region_id) VALUES (2, 'Ankobra River', 2) """)
conn.commit() # save again
```

STEP 4: Pull a single table into pandas

```
df_regions = pd.read_sql("SELECT * FROM regions", conn)
print(df_regions)
```

Output — a normal DataFrame, exactly like reading a CSV:

```
region_id region_name population
```

```
0      1    Ashanti   5000000
```

```
1      2    Western   2800000
```

From this point it's just pandas. `.info()`, `.describe()`, `.plot()` — everything you already know works exactly the same.

STEP 5: The power move — joining two tables in the query

This is where SQL is more powerful than plain CSV. Instead of merging DataFrames manually, you join inside the SQL query before pulling into pandas:

```
query = """
    SELECT
        rivers.river_name,
        regions.region_name,
        regions.population
    FROM rivers
    JOIN regions
    ON rivers.region_id = regions.region_id """
```

```
df_joined = pd.read_sql(query, conn)
print(df_joined)
```

Output:

```
river_name region_name population
```

```
0    Pra River    Ashanti   5000000
```

```
1 Ankobra River    Western   2800000
```

One clean DataFrame combining both tables. No manual merging in pandas needed — SQL did it before the data even arrived.

STEP 6: Close the connection

```
conn.close()
```

Always close when you're done. Think of it like closing a file after reading it

Full workflow summary in plain words:

galamsey.db file → your filing cabinet
conn = sqlite3.connect() → open the filing cabinet door
CREATE TABLE → add folders inside
INSERT INTO → put papers into the folders
pd.read_sql() → take papers out as a DataFrame
conn.close() → close the filing cabinet

25 SQL Tutorial 1 OPTIONAL / FUTURE — *Mode Analytics*

(Skip for now. Revisit only if a future project needs deep SQL-for-analytics work.)

<https://mode.com/sql-tutorial/>

2.6 SQL Zoo · PRACTICE ONLY — *SQL Zoo*

(Not a primary learning resource — use periodically to drill and retain SQL skills after finishing DataCamp)

https://sqlzoo.net/wiki/SQL_Tutorial

2.7 PostgreSQL Basics — *YouTube · optional*

<https://www.youtube.com/watch?v=qusjTv8N1cA>

PHASE 3 Machine Learning + WQU DS Projects

Add-On: Math Companion (paired with GFG Weeks 5-10)

✅ **Where it fits:** Foundation (watch before Week 5, ~3-4 hours total) These give you the visual "why" in ~3 hours. Don't skip these — everything below builds on them.

Resource	Covers	Link
3Blue1Brown — Essence of Linear	Vectors, dot product, matrices,	youtube.com/playlist?

Algebra	eigenvectors (intuition)	list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab
3Blue1Brown — Essence of Calculus	Derivatives, gradients, chain rule (intuition)	youtube.com/playlist?list=PLZHQObOWTQDMsr9K-rj53DwVRMYO3t5Yr

Week-by-week companion (the missing piece — StatQuest by Josh Starmer)

This is the resource that actually explains the math **behind each specific algorithm** you're learning, not generic math theory. Free YouTube, beginner-friendly, no jargon.

GFG Week	Day	Algorithm	StatQuest video to pair with it
Week 5	Day 30-31	Linear Regression, Gradient Descent	Linear Regression, Clearly Explained" + "Gradient Descent, Step-by-Step"
Week 5	Day 35	Logistic Regression	"Logistic Regression, Clearly Explained"
Week 6	Day 36	Model Evaluation, Confusion Matrix	"Sensitivity and Specificity, Clearly Explained"
Week 6	Day 38-39	Decision Trees	"Decision Trees, Clearly Explained"
Week 6	Day 41-42	SVM	"Support Vector Machines, Clearly Explained"
Week 7	Day 43	KNN	"StatQuest: K-nearest neighbors"
Week 7	Day 44-45	Naive Bayes	"Naive Bayes, Clearly Explained"
Week 7-8	Day 47-52	Ensemble, Boosting, Gradient Boosting	"Random Forests" + "Gradient Boost, Part 1-4"
Week 9	Day 58	K-Means, Clustering	"K-means clustering"
Week 10	Day 63-64	PCA, Dimensionality Reduction	"PCA, Step-by-Step" + "StatQuest: PCA main ideas"

Channel: youtube.com/@statquest — search each title, they're all free and 5-20 min each.

✅ If you want it more structured/formal

This roadmap already lists **Mathematics for Machine Learning Specialisation (Imperial College, Coursera audit-free)**. This is the one course that ties **linear algebra + calculus + PCA directly to ML**, with the free companion book at mml-book.com. Use this only if StatQuest + 3Blue1Brown leaves gaps (most people don't need it)

Bottom line: 3Blue1Brown first (one-time, ~3hrs) → then StatQuest video before each new algorithm as you hit it in GFG. No separate math phase needed.

✅ Fully covered by GFG Weeks 5–10. No external course required.

3.1 ML Basics + Regression — GFG Week 5

3.2 Model Evaluation + SHAP / Model Explainability + Classification — GFG Week 6 + SHAP

Do SHAP / Model Explainability immediately after Day 36. It takes one afternoon and you'll use it in every project after.

3.2.1 SHAP / Model Explainability — <https://shap.readthedocs.io/en/latest/>

3.2.2 MLflow Quickstart (NEW) — <https://mlflow.org/docs/latest/ml/getting-started/>

Install: pip install mlflow

Add these 3 lines to your Day 36 model and every model after:

```
import mlflow
mlflow.set_experiment('my-first-model')
with mlflow.start_run():
    mlflow.log_param('learning_rate', 0.01)
    mlflow.log_metric('accuracy', accuracy)
    mlflow.sklearn.log_model(model, 'model')
```

3.3 ML Algorithms — Decision Trees, SVM, KNN, Naive Bayes — GFG Week 7

3.4 Ensemble Learning — XGBoost, LightGBM, CatBoost — GFG Week 8

3.5 Unsupervised Learning — Clustering, GMM, DBSCAN — GFG Week 9

3.6 Dimensionality Reduction + Deployment Introduction — GFG Week 10

3 b #1 — Stanford CS229: Machine Learning (Andrew Ng)

✅ **Where it fits:** Complete after GFG Weeks 5–10. Watch CS229 as a companion or immediately after to solidify the math behind everything GFG taught you.

Stanford University · Full Graduate Course · 27+ hours · Free on YouTube 🔗

<https://www.youtube.com/playlist?list=PLoROMvodv4rMiGQp3WXShMGgzqpVfbU>

This is the one. Andrew Ng's full Stanford graduate-level ML course — not the Coursera version. This is the original, with full mathematical derivations. Every algorithm you touched in GFG (regression, SVMs, decision trees, EM, RL) is explained from first principles here. Lecture notes alone are some of the best ML reading material ever written.

What you will learn:

- Linear/logistic regression with full mathematical derivation
- Generative vs discriminative models — GDA, Naive Bayes
- SVMs, kernels, and the full dual formulation
- Decision trees, ensemble methods, boosting
- EM algorithm and mixture of Gaussians
- PCA, ICA, factor analysis
- Reinforcement Learning — MDPs, value iteration, policy gradient
- Debugging ML models and error analysis (lecture 13 alone is worth everything)

3 c — WQU Data Science Projects

→ **Applied Data Science Lab**

<https://www.wqu.edu/adsl>

PHASE 4 Git · Docker · CI/CD · Core MLOps

⚠ GFG does not teach these tools. Complete before moving to Deep Learning.

4.1 — Git & GitHub

- **Git & GitHub Full Course** — *freeCodeCamp*
<https://www.freecodecamp.org/news/git-and-github-crash-course-for-beginners/>
- **Microsoft Learn for GitHub** — *Microsoft*
<https://learn.microsoft.com/en-us/training/github/>

4.2 — Docker

- **Docker Tutorial for Beginners** — *TechWorld with Nana · YouTube*
<https://www.youtube.com/watch?v=3c-iBn73dDE>
- **Docker for Beginners** — *freeCodeCamp*
<https://www.freecodecamp.org/news/an-introduction-to-docker-and-containers-for-beginners/>

4.3 — CI/CD

- **GitHub Actions Full Course** — *YouTube*
<https://www.youtube.com/watch?v=mFFXuXjVgkU>

4.4 — Core MLOps

- **Made With ML — MLOps** — *madewithml.com · best free MLOps resource available · primary*
<https://madewithml.com>
- **MLOps Zoomcamp** — *DataTalksClub · alternative / highly practical*
<https://datatalks.club/blog/mlops-zoomcamp.html>

PHASE 5 Deep Learning + Projects + Reinforcement Learning + DSA + DE

✅ Fully covered by GFG Weeks 11–16. No external course required.

5.1 Introduction to Deep Learning — GFG Week 11

5.1.1 Weights & Biases (NEW) — <https://wandb.ai/site/courses/101/>

Do this in the same week as Week 11, before your first neural network finishes training.

1. **Step 1:** Check access first — wandb.ai may have geographic restrictions affecting Ghana since September 2025. Test it before relying on it.
2. **Step 2:** If accessible — W&B 101 free course (~30 min): <https://wandb.ai/site/courses/101/>
3. **Step 3:** From this point forward, every single model you train for the REST of Phase 5 — Week 12 through David Silver's RL course at the very end — gets logged in W&B if accessible, or MLflow (from Phase 3) if W&B is not accessible. You don't add W&B again at any other step below. You just keep using it.

5.2 Optimisation + MLP + Backpropagation — GFG Week 12

5.3 CNN + RNN + Time Series — GFG Week 13

5.4 LSTM, GRU, Transformers — GFG Week 14

5.5 GANs + Autoencoders — GFG Week 15

✓ GFG backup

5.6 Practical Deep Learning *fast.ai*

<https://course.fast.ai/>

5.5 Dive into Deep Learning

<https://d2l.ai/>

5.2 Andrej Karpathy: Neural Networks — Zero to Hero

Where it fits: Add these alongside or immediately after GFG Weeks 11–16. MIT 6.S191 is your reinforcement layer. Karpathy is your deep understanding layer.

Former Tesla AI Director, ex-OpenAI · Full YouTube Series · Free 🔗

<https://www.youtube.com/playlist?list=PLAqhlrjkxbuWI23v9cThsA9GvCAUhRvKZ>

This is the most important deep learning resource on the internet right now. Karpathy builds everything from absolute scratch — backpropagation, micrograd, character-level language models, transformers, GPT — line by line in Python. No black boxes. By the end you have built GPT yourself and understand every line of it.

What you will learn:

- Backpropagation from scratch — building micrograd (your own autograd engine)
- makemore — building a character-level language model from scratch
- MLPs, batch normalisation, weight initialisation — built from scratch
- WaveNet-style architectures
- "Let's build GPT from scratch" — the single most important video
- Tokenisation deep dive
- "Let's reproduce GPT-2" — building a 124M parameter GPT-2 from scratch

Why this is non-negotiable: Karpathy has the rare ability to make complex things genuinely simple. His "Let's build GPT" video alone has been called the best educational content on AI ever made. This is what separates engineers who use LLMs from engineers who understand them

5.3 #1 — MIT 6.S191: Introduction to Deep Learning (Alexander Amini)

Where it fits: Add these alongside or immediately after GFG Weeks 11–16. MIT 6.S191 is your reinforcement layer. Karpathy is your deep understanding layer.

MIT · Full Course · Updated Every Year (2026 Edition Now Live) · Free on YouTube

Course: https://www.youtube.com/playlist?list=PLtBw6njQRU-rwp5_7C0oIVt26ZgjG9NI

Lab: <https://introtodeeplearning.com/>

MIT's official deep learning course, updated annually and completely free. Unlike older courses, this covers modern topics — transformers, LLMs, diffusion models, RL — in a clean, structured way. Alexander Amini is an exceptional teacher. Every lecture has accompanying Colab notebooks you can run for free.

What you will learn:

- Deep learning foundations — neurons, backpropagation, optimisation
- CNNs — architecture, visualisation, transfer learning
- RNNs, Transformers, and Attention
- Deep Generative Modelling — VAEs, GANs, Diffusion
- Reinforcement Learning (dedicated lecture)
- Language Models and New Frontiers (dedicated lecture — covers LLMs)

- AI in biology, robotics, and science

5.6 — WQU Deep Learning Projects

→ Deep Learning Fundamentals Lab

<https://www.wqu.edu/deep-learning-lab>

5.7 a— Core Reinforcement Learning

5.7.1 Reinforcement Learning — *GFG Week 16*

5.7.2 Deep Learning — *Hugging Face Deep RL Course*

<https://huggingface.co/learn/deep-rl-course/unit0/introduction>

5.7.2 PPO (Proximal Policy Optimization) — *Hugging Face Deep RL Course*

<https://huggingface.co/learn/deep-rl-course/unit8/introduction>

5.7 b— David Silver: RL Course (UCL / DeepMind)

Where it fits: Do after GFG Week 16 and the Hugging Face Deep RL Course already in your roadmap. David Silver goes deeper into theory.

University College London + DeepMind · 10 Lectures · Free on YouTube

<https://www.youtube.com/playlist?list=PLqYmG7hTraZDM-OYHWgPebj2MfCFzFObQ>

David Silver invented AlphaGo. He is one of the top 3 RL researchers alive. This 10-lecture course is the mathematical foundation of modern RL — every paper in the field builds on the concepts taught here. It is older (2015) but the theory has not changed.

What you will learn:

- Markov Decision Processes — the mathematical framework for RL
- Dynamic Programming — value iteration, policy iteration
- Model-free prediction — Monte Carlo, TD learning
- Model-free control — SARSA, Q-learning

- Value function approximation — neural networks in RL
- Policy gradient methods — the foundation of PPO and TRPO
- Integrating learning and planning — Dyna, MCTS
- Exploration vs exploitation — the core RL tradeoff
- Case study: RL in classic games (foundation of AlphaGo)

5.8 — DSA

5.8.1 Problem Solving with Algorithms and Data Structures using Python — *Runestone Academy*

<https://runestone.academy/ns/books/published/pythonds/index.html>

5.8.2 DSA (Video lectures) — *freeCodeCamp*

<https://www.youtube.com/watch?v=pkYVomU3MgA>

5.8.3 Practice Problems — *NeetCode 150*

<https://neetcode.io/practice>

ML-specific DSA topics to focus on inside NeetCode 150:

Arrays and hashing	Feature engineering, data manipulation
Binary search	Efficient model search, hyperparameter tuning
Trees	Decision tree implementation questions
Graphs	Graph neural networks, knowledge graphs
Dynamic programming	Sequence models, Viterbi algorithm
Sliding window	Time series, sequence processing

Skip: competitive programming, segment trees, Fenwick trees, advanced graph algorithms. None of these appear in ML engineering interviews.

Companion Document

Phase 6 has a dedicated full course guide covering all topics below in depth, with the top 3 free comprehensive courses per subject, structured in the exact order you should study them.

→ **See: Modern AI Engineering — Full Course Guide**

Topics — in learning order

- 6.1 Natural Language Processing** — *Transformers, BERT, GPT, text classification, NER, summarisation*
- 6.2 Computer Vision** — *CNNs, object detection, segmentation, OpenCV, real-time applications*
- 6.3 Large Language Models (LLMs)** — *Architecture, training dynamics, fine-tuning, deployment*
- 6.4 Generative AI** — *RLHF, diffusion models, full-stack GenAI engineering*
- 6.5 RAG — Retrieval-Augmented Generation** — *Building, evaluating, and optimising RAG pipelines*
- 6.6 Vector Databases** — *Embeddings, ANN search, ChromaDB, Weaviate, Pinecone, hybrid search*
- 6.7 Agentic AI** — *Agents, tool use, multi-agent systems, LangGraph, AutoGen*
- 6.8 Prompt Engineering + Fine-Tuning** — *Prompt design, LoRA, QLoRA, PEFT, instruction tuning*
- 6.9 FastAPI for AI Services** — *Building and deploying AI-powered APIs*

PHASE 7 Advanced Track — Cloud · Certification

🔴 Complete Phases 5 and 6 before starting Phase 7. These topics require full deep learning and AI engineering knowledge to use meaningfully.

7.1 — Advanced MLOps at Scale

→ **MLOps Zoomcamp** — *DataTalksClub · complete if not done in Phase 4*
<https://datatalks.club/blog/mlops-zoomcamp.html>

7.2 — AWS for ML Engineers

Step 1 — AWS SageMaker Studio Lab (free · no AWS account · no credit card)

<https://studiolab.sagemaker.aws/>

- Free JupyterLab environment with CPU and GPU access
- 15 GB of persistent storage — files survive after session ends
- Sign up with just your email. No billing. No credit card. Ever.
- Practice real SageMaker workflows with zero financial risk

Step 2 — AWS Skill Builder: Machine Learning Learning Plan (free tier)

https://explore.skillbuilder.aws/learn/learning_plan/view/28/machine-learning-learning-plan

- Official AWS ML learning path — free tier covers all foundations
- Translates your Phase 3 classical ML (CS229, GFG Weeks 5–10) into SageMaker
- Prepares you for AWS Machine Learning Specialty if you choose to pursue it

Step 3 — AWS Machine Learning University: Accelerated Tabular Data (free · official AWS · GitHub)

<https://github.com/aws-samples/aws-machine-learning-university-accelerated-tab>

- The exact ML course used to train Amazon's own developers, made public
- Hands-on notebooks for tabular data — the same problem type as your Phase 3 ensemble learning week
- Skip MLU's NLP and Computer Vision courses — you already have CS224N and CS231n, both deeper than AWS's versions

Step 4 — AWS Skill Builder: Generative AI Learning Plan for Developers (free · 5 courses · 11 hours) ★ MOST RELEVANT

https://explore.skillbuilder.aws/learn/learning_plan/view/2068/generative-ai-learning-plan-for-developers

- Introduction to Generative AI — Art of the Possible
- Planning a Generative AI Project
- Amazon Bedrock — Getting Started (fundamental)
- Foundations of Prompt Engineering (intermediate)
- Building Generative AI Applications using Amazon Bedrock (advanced)
- Directly matches your Phase 6 LLM/RAG/Agents work — shows you how LangChain concepts you already know plug into AWS's managed LLM service (Bedrock) instead of calling OpenAI/Anthropic APIs directly

Step 5 — AWS Skill Builder: Generative AI Learning Plan for Model Builders (conceptual only — no real hands-on access to HyperPod)

<https://skillbuilder.aws/learning-plan/BB4UWE24NX/generative-ai-learning-plan-for-model-builders/1DUA79ANG5>

- Amazon SageMaker JumpStart Foundations
- Customizing and Evaluating LLMs using Amazon SageMaker JumpStart
- Building Language Models on AWS — SageMaker HyperPod
- Direct AWS equivalent of your Phase 6.8 fine-tuning work (LoRA/QLoRA/PEFT), but at infrastructure scale instead of a free Colab/Kaggle GPU
- Only do this if you want to go deep on fine-tuning at AWS scale

Step 6 — AWS Skill Builder: Financial Services Learning Plan (free · 30 min)

<https://skillbuilder.aws/learning-plan/S7NV16QY54/financial-services-learning-plan-artificial-intelligence-and-machine-learning/JEV4EHJERR>

- Covers managing data with Amazon SageMaker and running AI/ML solutions like text classification and image recognition using Comprehend, Rekognition, and Transcribe
- Fintech and mobile money (M-Pesa, MTN Mobile Money) are among the biggest AI/ML opportunity areas on the continent; fraud detection and credit scoring for people with no traditional credit history
- Only 30 minutes — low cost, real differentiation for your goal of becoming Africa's greatest AI/ML engineer

7.3 — Microsoft Azure ML (new) Microsoft Learn: Azure ML Path — completely free, no credit card

<https://learn.microsoft.com/en-us/training/paths/create-machine-learn-models/>

- Official free Azure ML training from Microsoft
- Free sandbox environments — practice without any billing
- Covers Azure ML Studio, pipelines, and AutoML

7.4— Google Professional Machine Learning Engineer ★ Capstone

The Google Professional ML Engineer is one of the most respected cloud ML credentials globally. Attempt only after completing all phases above.

Step 1 Google Cloud Skills Boost — ML Path — *Official pre-certification curriculum — complete first*
<https://www.cloudskillsboost.google/paths/17>

Step 2 Official Exam Guide — *Free · everything testable is listed here*
<https://cloud.google.com/learn/certification/machine-learning-engineer>

Step 3 Preparing for the Professional ML Engineer Exam — *Google + Coursera · audit free · official exam prep*
<https://www.coursera.org/learn/preparing-cloud-professional-ml-engineer-exam>

Step 4 Hands-on Vertex AI Labs — *Google Cloud Skills Boost · complete every ML quest*
<https://www.cloudskillsboost.google/>

7.C — Data Engineering Track (optional)

Data Engineering Zoomcamp — DataTalksClub

<https://datatalks.club/blog/data-engineering-zoomcamp.html>

Only pursue this if you specifically want to go deep on pipelines — Kafka, Spark, dbt, BigQuery, Airflow.

Portfolio Strategy — 5 Projects You Must Have

🔴 These are the 5 projects every ML Engineer portfolio needs — build them at the phase indicated, not all at the end.

1	EDA + ML Prediction Project	Phase 3	Data cleaning, feature engineering, model evaluation, SHAP explanations
2	Dockerized ML API (FastAPI + Docker)	Phase 4	MLOps, deployment, CI/CD, reproducibility
3	Deep Learning Project (image or text)	Phase 5	Neural networks, GPU training, transfer learning
4	RAG Application (LLM + Vector DB)	Phase 6	LLMs, retrieval, production AI engineering
5	Agentic AI System (multi-agent + tool use)	Phase 6	LangGraph, agents, production deployment

Every GitHub repository must have:

- README with: what it does, what problem it solves, tech stack, how to run, results/metrics
- A demo GIF or screenshot at the top
- requirements.txt or environment.yml
- Inline code comments

✦ *Bonus for Anthropic-track candidates: Publish your ARENA mech interp mini-project on GitHub with a clear README, and write a short post on the Alignment Forum explaining what you found. A small honest experiment published publicly is worth more than any credential in the safety community.*

PHASE 8 Job Search and Career Preparation

✦ Phase 8 runs parallel with Phase 7, not after it. Start when you have at least 2–3 portfolio projects live on GitHub.

8.1 — CV / Resume

[Jake's Resume Template](#) — the most widely used one-page tech resume · Free

<https://www.overleaf.com/latex/templates/jakes-resume/syzfjbzwjnsc>

For ML Engineer roles, your CV sections:

- Education — degree, relevant courses
- Skills — Python · PyTorch · SQL · Docker · MLflow · LangChain · LangGraph · FastAPI
- Projects — 3–5 projects with metrics, not just descriptions
- Experience — relevant work, internships, freelance
- Certifications — WQU, Google ML Engineer, BlueDot (Anthropic track)

✦ *Critical rule: Every project bullet must have a number. Not 'built a RAG application' — 'built a RAG application achieving 87% retrieval accuracy on a 10,000-document corpus with sub-200ms latency.' Numbers are what get you past screening.*

8.2 — ML Technical Interview Prep

ML Theory Interviews

Chip Huyen — Introduction to Machine Learning Interviews (free book)

<https://huyenchip.com/ml-interviews-book/>

- 200+ questions with detailed answers
- Covers ML theory, data, coding, and system design
- Written by Stanford lecturer and ex-NVIDIA engineer

ML System Design Interviews

Designing Machine Learning Systems — Chip Huyen <https://www.oreilly.com/library/view/designing-machine-learning/9781098107956/>

- Standard reference for ML system design interviews
- Given a problem, you walk through: framing, data, features, model, evaluation, deployment, monitoring
- Combine with your existing 6.10 ByteByteGo and Exponent resources

8.3 — ML Take-Home Assessments

The structure that wins:

- EDA first — always. Show you understand the data before touching models.
- Baseline model — start simple (logistic regression or XGBoost). Document it.
- Iterate — improve from baseline with clear reasoning. Each step logged with Mlflow.
- Evaluation — always report multiple metrics. Use SHAP.
- Write-up — treat the README like a report. Explain every decision.

What separates good from great:

- Honest about limitations — 'model performs poorly on class X due to data imbalance'
- Reproducible — requirements.txt, random seeds set, clear run instructions
- Production-aware — 'in production I would add monitoring for data drift'

8.4 — Where to Apply

LinkedIn	Filter: Machine Learning Engineer, AI Engineer, MLOps Engineer
Wellfound (AngelList)	Startup ML roles — less competition than big tech
levels.fy	Compensation data for ML roles globally — know your worth
Otta	Curated tech job board, strong for international applicants

Build in public — the strategy that gets you noticed:

- **Post every project on LinkedIn with a short explanation of what you built and learned**
- **Write one short technical post per month — 'What I learned building a RAG system from scratch'**
- **Kaggle — even a bronze medal is visible and many recruiters check**
- **GitHub stars — if your ARENA or RAG project is useful, people will star it**

GeeksForGeeks Course (all GFG phases): <https://www.geeksforgeeks.org/batch/ds-16?tab=Resources>
DeepLearning.AI Short Courses (Phase 6 permanent resource): deeplearning.ai/short-courses
GPU Access — Kaggle Notebooks (Free T4 GPU · 30hrs/week · use for Phase 5, 6.8, and ARENA):
www.kaggle.com/code
GPU Access — Google Colab (Free T4 GPU backup · use when Kaggle quota runs out ·):
<https://colab.research.google.com/>